

Feasibility Study

1. Technical Feasibility

Assessment:

- **Backend:** ASP.NET Core Web API with Entity Framework Core and Microsoft SQL Server provides a high-performance, robust, and secure infrastructure for managing transactional financial data, role-based access control (RBAC), and JWT authentication.
- **Frontend:** React.js, HTML, CSS, and JavaScript offer a modular, responsive, and dynamic user interface suitable for both desktop and mobile web platforms
- **Real-time & Document Generation:** ASP.NET Core SignalR provides seamless real-time messaging, while document generation tools (e.g., QuestPDF) allow instant generation of digital passbooks, receipts, and credential summaries.

Infrastructure:

- The decoupled architecture (Web API + Single Page Application) allows independent scaling of backend services and frontend delivery.
- Cloud hosting on services like Microsoft Azure or AWS can effortlessly scale the system from a single Panchayath to district-level or state-wide Kudumbashree networks.

Challenges:

- **Data Integrity & Dynamic Calculations:** Managing concurrent weekly transactions and dynamic reducing interest rate calculations across multiple micro-loans requires strict database transaction locks and robust backend validation.
- **Rural Connectivity:** Ensuring smooth usability for local Ayalkoottam members who may experience unstable mobile network coverage in rural areas.

Conclusion:

The project is technically feasible. The selected Microsoft .NET and React stack is industry-standard, offering high performance, security, and scalability necessary for real-time micro-finance tracking.

2. Economic Feasibility

Assessment:

- **Initial Costs:**
 - Software Stack: Built on open-source frameworks (React, ASP.NET Core) and standard SQL database infrastructure, minimizing initial software licensing overhead.
 - Infrastructure: Minimal initial hosting requirements as a prototype web application, with predictable cloud hosting scaling costs as adoption grows.
- **Cost Savings for Users:**
 - Elimination of Paper Operations: Replaces physical registers, printed passbooks, and manual receipts with immutable digital ledgers and downloadable PDF reports.
 - Reduction of Financial Anomalies: Automated calculations for thrift savings collections, reducing interest loans, and penalties eliminate costly manual accounting errors and auditing overhead.

Conclusion:

The project is economically feasible. The automation of ledger entries and loan lifecycles delivers clear operational savings that outweigh development and hosting expenses.

3. Operational Feasibility

Assessment:

- **Platform Users:**
- **CDS Authorized Personnel:** Streamlined administrative control to register new Ayalkoottam units, manage ward details, and assign office bearers.
- **Office Bearers (President, Secretary & Treasurer):** Specialized workflows to schedule weekly meetings, record attendance, enter thrift deposits, approve loan requests, and disburse EMIs.
- **Members:** Simplified self-service portal to monitor personal savings, view attendance history, track loan application status, and engage in unit messaging.

- **User Training:**

- The step-by-step onboarding process—where temporary credentials are administrative outputs converted to PDFs and forced password changes occur on initial login—ensures smooth transition and digital security for rural users.

Challenges:

- Digital Literacy: Varying levels of technological familiarity among rural Ayalkoottam members require an intuitive UI with plain-language inputs and visual indicators.
- Change Management: Shifting local office bearers from physical ledger books to real-time digital entry during weekly meetings.

Conclusion:

The project is **operationally feasible**. By directly mirroring the official Kudumbashree governance model and role hierarchy, SAHAYI minimizes workflow disruption while significantly improving governance and transparency.